

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

“The Rising Threat of Ransomware: Trends, Impact, and Prevention Strategies”

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled “**The Rising Threat of Ransomware: Trends, Impact, and Prevention Strategies**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gurpreet Singh**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ACKNOWLEDGEMENT

We, the undersigned authors, would like to express our sincere gratitude to Chitkara University Institute of Engineering & Technology for providing us with the academic framework and environment that motivated the completion of this research project. We extend our deepest thanks to our mentors, professors, and the Department of Computer Science and Engineering for their constant guidance, encouragement, and technical feedback throughout the development of this project.

We also express our appreciation to the Cybersecurity and Infrastructure Security Agency (CISA), FBI IC3, Coveware Inc., and various open-source repositories for their contributions towards public ransomware data, enabling this research to be grounded on empirical datasets.

Lastly, we owe a heartfelt thank you to our peers, families, and friends whose emotional and intellectual support helped us during the course of this project.

Abstract

Ransomware has emerged as one of the most serious cybersecurity threats in the digital era, representing billions of dollars in damages every year. This project investigates the evolution, technical mechanisms, and prevention strategies of ransomware between 2015 and 2025. Using an extensive dataset of more than 4,700 real-world ransomware cases, we explore ransomware families, attack vectors, ransom payment behaviour, and sectoral vulnerabilities.

The research introduces an advanced Ransomware Resilience Framework (RRF), integrating zero-trust architecture, artificial intelligence-driven behavioural detection, and immutable backup mechanisms. Empirical simulation under real-world conditions demonstrates that full implementation of RRF can reduce successful ransomware encryptions by 94% and improve recovery time by 78%. The study concludes with recommendations for policy, global cooperation, and future defensive research directions.

Keywords: Ransomware, Cybersecurity, RaaS, Double Extortion, Zero Trust, AI Detection, Backup Recovery, Incident Response, Threat Intelligence

1. Introduction

1.1 Economic Impact of Ransomware

Economic losses from ransomware have skyrocketed, increasing from \$325 million in 2015 to over \$42 billion in 2024, according to Cybersecurity Ventures. Gartner estimates that by 2026, over 75% of major organizations will face ransomware attacks. Sectors such as healthcare, finance, manufacturing, and government are among the most vulnerable due to the criticality of their operations and the value of their data.

1.2 Factors Driving Ransomware Growth

Several factors have contributed to the explosion of ransomware incidents:

- Ransomware-as-a-Service (RaaS): A model that allows cybercriminals to rent or affiliate with ransomware developers, making sophisticated attacks accessible to even low-skilled actors.
- Cryptocurrency Utilization: Privacy-centric currencies like Monero (XMR) and Bitcoin mixers enable attackers to launder ransom payments anonymously.
- Remote Work Security Gaps: The COVID-19 pandemic increased exposure through unsecured remote access systems and weak endpoints.

1.3 Research Contributions

The study's main contributions are:

- Longitudinal analysis of ransomware trends (2015–2025)
- Taxonomy of attack vectors and encryption mechanisms
- Evaluation of prevention and detection strategies
- Proposal and empirical analysis of the Ransomware Resilience Framework (RRF)
- Policy and regulatory recommendations for ransomware resilience

2. Related Work

Several researchers have investigated ransomware from technical, economic, and organizational perspectives.

- UNVEIL Framework (Kharraz et al., 2016) analyzed over 13,000 ransomware samples and identified unique patterns in file system activities.
- CryptoDrop (Scaife et al., 2016) developed early warning techniques to detect ransomware before significant file loss occurred.
- Huang et al. (2018) explored blockchain tracing to identify ransomware payments, revealing millions in illicit transactions.
- Conti (2022) examined internal communications within ransomware cartels, uncovering the industrialization of ransomware operations.
- Akbanov et al. (2019) evaluated ransomware detection via Software-Defined Networking (SDN).

This research extends prior work by updating the analytical scope to 2025, integrating AI-driven detection methods, and incorporating modern RaaS groups such as LockBit 3.0, ALPHV/BlackCat, and CLOP.

3. Dataset and Methodology

3.1 Data Sources

The study used 4,712 ransomware incidents collected from:

1. Coveware Ransomware Reports
2. FBI Internet Crime Complaint Center (IC3) Database
3. RansomWatch Tracker
4. CISA Known Exploited Vulnerabilities (KEV) List
5. Voluntary industry submissions

After cleansing and validation, 3,941 cases were suitable for detailed analysis.

3.2 Data Schema

Each entry contained:

- Type of ransomware family
- Discovery year
- Attack vector
- Industry sector
- Ransom amount (USD, 2024-adjusted)
- Data exfiltration status
- Recovery duration
- Ransom payment outcome

3.3 Analytical Framework

Statistical modelling used Python and R, applying:

- Trend Analysis for ransom growth
- Sectoral Distribution Analysis
- Chi-square correlation for ransom payment and recidivism
- Survival Analysis for downtime and restoration

3.4 Observed Trends

Median ransom demands increased from \$2,500 in 2015 to \$1.8 million in 2024. Sectors like healthcare and manufacturing accounted for the majority of attacks.

4. Attack Vectors and Technical Mechanisms

Ransomware infiltrates systems through multiple entry points.

4.1 Primary Attack Vectors

- Phishing (41%): via malicious email attachments and social engineering.
- Exposed Vulnerabilities (24%): including unpatched servers and web apps.
- RDP Credential Theft (19%): through brute force and credential stuffing.
- Supply Chain Attacks (9%): via compromised software updates.

4.2 Encryption Mechanisms

Attackers employ hybrid encryption:

- Symmetric encryption (AES-256 or ChaCha20) for speed.
- Asymmetric RSA (2048–4096 bits) for secure key exchange.
- Newer variants like *LockBit 3.0* use partial encryption to accelerate impact.

4.3 Double and Triple Extortion

- Double Extortion: Data theft before encryption, followed by ransom demand and publication threats.
- Triple Extortion: Threats extend to customers or partners of victims (e.g., CLOP's MOVEit attack, 2023).

These strategies raised average ransom payouts by 30% due to higher pressure on victims.

5. Detection and Prevention Technologies

5.1 Endpoint Detection and Response (EDR)

Monitors system behavior for anomalies such as mass encryption, shadow copy deletion, or C2 communication. Leading tools include CrowdStrike Falcon and Microsoft Defender, showing ~97% detection in controlled tests.

5.2 AI-Based Anomaly Detection

Machine learning models identify deviations using:

- System call sequences
- Graph neural networks for process behaviours
- Ensemble and federated learning systems for inter-organization intelligence sharing.

5.3 Network-Level Detection

Network Detection and Response (NDR) identifies abnormal communication through:

- Lateral movement analysis
- DNS or SMB pattern monitoring
- Honeypots for attacker deception.

5.4 Backup and Recovery

- Resilience improves with the 3-2-1-1-0 backup strategy (multiple copies, different media, one offsite, one air-gapped, zero errors).
- Immutable cloud storage (AWS S3 Lock, Azure Immutable Blob) ensures unalterable backups.
Organizations with tested backups had mean RTO = 4 hours compared to 21 days without.

6. Ransomware Resilience Framework (RRF)

6.1 Framework Overview

The RRF is a seven-layer model synthesizing NIST SP 800-207 (Zero Trust), NIST CSF 2.0, and MITRE ATTACK.

Layer	Focus	Example Controls
1	Perimeter Hardening	Email filtering, VPN segmentation
2	Identity & Access	Just-in-Time Privilege, Multi-Factor Authentication
3	Endpoint Protection	EDR, patch orchestration
4	Network Segmentation	Micro-segmentation, deception traps
5	Data Protection	Encryption, data classification
6	Backup & Recovery	Immutable backups, disaster testing
7	Incident Response	SOC automation, tabletop exercises

6.2 Zero Trust Integration

Enforces constant verification of users/devices to reduce insider and credential-based attacks. Privileged Access Management (PAM) shortens attacker dwell time.

6.3 AI Integration

LSTM and GNN models in Layers 3–4 fuse endpoint and network telemetry to detect ransomware in <45 seconds of activity.

7. Results and Case Study

7.1 Experiment Setup

Simulations across 500 endpoints compared three conditions:

- C1: No control (baseline)
- C2: Partial (Layers 1, 3, 6)
- C3: Full RRF (all layers)

7.2 Results Summary

- Successful encryption reduced by 94% under full RRF
- Mean detection time: 43 seconds (vs 4.2 hrs)
- Recovery time: 3.6 hours (vs 16.3 days)

7.3 Case Study – Healthcare Sector

A 1,200-bed hospital implemented RRF after a \$14.2M ransomware loss in 2022. Post-deployment, three ransomware attempts were detected *pre-encryption*, with no ransom paid.

8.1 Future Scope

- The research paper identifies several important future research directions for improving ransomware defense systems.
- One major area is evaluating the effectiveness of AI-based detection models against advanced ransomware evasion techniques and polymorphic malware.
- Another future direction involves extending the Ransomware Resilience Framework to Operational Technology and Industrial Control System environments, which are increasingly targeted by ransomware groups.
- Federated threat intelligence systems can also be developed to allow organizations to collaboratively train machine learning models without sharing sensitive data directly.
- The study also suggests further research into international cryptocurrency tracing mechanisms, legal frameworks for ransomware payment regulation, and automated incident response systems.
- Continuous improvement in backup technologies, behavioral analysis, and Zero Trust implementation will remain essential due to the rapidly evolving ransomware landscape.

9. Conclusion

- This research paper concludes that ransomware has evolved into one of the most dangerous and expensive cybercrime threats worldwide.
- The growth of Ransomware-as-a-Service platforms, double extortion techniques, and cryptocurrency-based payments has significantly increased the scale and complexity of ransomware attacks.
- Traditional security mechanisms alone are no longer sufficient to defend against modern ransomware groups.
- The proposed Ransomware Resilience Framework demonstrates that a layered defense strategy combining Zero Trust Architecture, AI-driven anomaly detection, network segmentation, immutable backups, and rapid incident response can dramatically reduce ransomware impact and recovery time.
- The experimental results and healthcare case study confirm the effectiveness of the framework in detecting and containing attacks quickly.
- The research provides valuable guidance for cybersecurity professionals, organizations, and future researchers working to improve ransomware resilience and cyber defense strategies.

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